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Original article

Linkages between life history type and migration pathways in freshwater and marine environments for Chinook salmon, *Oncorhynchus tshawytscha*

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ABSTRACT

Chinook salmon, *Oncorhynchus tshawytscha*, are commonly categorized as ocean-type (migrating to the ocean in their first year of life) or stream-type (migrating after a full year in freshwater). These two forms have been hypothesized to display different ocean migration pathways; the former are hypothesized to migrate primarily on the continental shelf whereas the latter are hypothesized to migrate off the shelf to the open ocean. These differences in migration patterns have important implications for management, as fishing mortality rates are strongly influenced by ocean migration. Ocean-type Chinook salmon predominate in coastal rivers in the southern part of the species' range, whereas stream-type predominate in the interior and northerly rivers. This latitudinal gradient has confounded previous efforts to test the hypothesis regarding ocean migration pathways. To address this problem, we used a pair-wise design based on coded wire tagging data to compare the marine distributions of stream- and ocean-type Chinook salmon from a suite of rivers producing both forms. Both forms of Chinook salmon from the lower Columbia River, Oregon coast, lower Fraser River, and northern British Columbia rivers followed similar migration paths, contradicting the hypothesis. In contrast, recoveries of tagged Chinook salmon from the upper Columbia River, Snake River, and the upper Fraser River revealed migration patterns consistent with the hypothesis. These findings have important implications for our understanding of these life history types, and also for the conservation and management of declining, threatened, or endangered stream-type Chinook salmon populations in the US and Canada.
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1. Introduction

The conservation of migratory animals depends on maintaining quality habitat and minimizing exploitation in each of the regions that they occupy, and passage between all necessary habitats. Migrations represent complex combinations of evolved and plastic responses to environmental conditions (Swingland and Greenwood, 1984; Dingle, 1996), complicating our understanding of the phenomenon. The world's fisheries predominately exploit migratory species, so temporal and spatial distribution patterns are critical for conservation (Jennings et al., 2001). For anadromous fishes, the great majority of total growth is achieved at sea after having left less productive freshwater environments (Gross, 1987; McDowall, 1988). There are complex linkages between the life history patterns in these two environments, including the size, age, and timing of seaward migration, the duration of marine residence, and the size and timing of return migration (Quinn et al., 2009). Upon entering marine waters, juveniles can occupy a wide range of estuarine, nearshore, shelf, and open ocean habitats. Among the species of Pacific salmon and trout (*Oncorhynchus*, the best studied group), there is considerable variation in the spatial extent and duration of marine migrations (Quinn and Myers, 2004).

Perhaps the most complex combination of life history and migration patterns is exhibited by Chinook salmon, *Oncorhynchus tshawytscha*. Towards the southern end of their range, and especially in low elevations, juveniles typically migrate to sea in their first spring or summer of life. These were termed "sea-type" by Gilbert (1913) on the basis of scale pattern analysis. More recently, these Chinook salmon have been referred to as ocean-type, and this pattern is associated with relatively early emergence and rapid growth in freshwater (reviewed by Taylor, 1990a, 1990b; Healey, 1991). In higher elevations within the southern part of the range, and in all cases in the northern part of the range, Chinook salmon spend a full year in freshwater before migrating to sea at a larger size, as "stream-type" fish. Populations north of about 56°N latitude are virtually all dominated by or exclusively stream-type

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individuals (Taylor, 1990a). Healey (1991) argued that stream-type and ocean-type Chinook salmon represent distinct "races", characterized by profound differences in many aspects of their behavior and ecology (Table 1), whereas Brannon et al. (2004) emphasized the importance of environmental conditions (especially temperature) in determining the life history and behavior of the two types. Evidence pertaining to this controversy is mixed. The forms differ in behavior and morphology as juveniles (Taylor, 1990b), but genetic analysis by Waples et al. (2004) indicated that in most cases the two types diverged independently in many different rivers (i.e., are polyphyletic). We use the term "type" rather than "race" when referring to these life history variants for consistency with the common and long-standing use of "stream-type" and "ocean-type" but this does not imply a judgment regarding the evolutionary status of these salmon. However, the interior Columbia River basin populations seem to represent a lineage distinct from coastal populations (Waples et al., 2004).

Stream-type populations, especially in the southern part of their range, often return in spring, whereas ocean-type populations typically return in fall, though both types spawn in fall (Healey, 1991; Myers et al., 1998; Waples et al., 2004). Moreover, Healey (1983) and Healey and Groot (1987) pointed out that ocean-type Chinook salmon seem to be distributed primarily along the continental shelf during their marine residence whereas stream-type Chinook salmon are more common off the shelf and on the open ocean. The ocean migration patterns of these two types of Chinook salmon profoundly affect their conservation and management. Salmon fishing on the open ocean is negligible since the elimination of the high seas fisheries in international waters. Thus chum, pink, and sockeye salmon are almost exclusively taken as maturing fish on their homeward migration in coastal waters. In contrast, Chinook salmon are vulnerable for several years (i.e., the period they spend at sea: Roni and Quinn, 1995) if they remain in coastal waters, as many do. The tendency to migrate into offshore waters to feed would greatly reduce their vulnerability to coastal fisheries, as noted by Waples et al. (2004). Many Evolutionarily Significant Units (ESU) of Chinook salmon have been designated as threatened or endangered under the United States Endangered Species Act (Good et al., 2005). Improved understanding of Chinook migration and exploitation patterns would greatly assist conservation efforts, as many of these population complexes are still exploited in commercial and recreational fisheries.

Despite the time, effort, and money that have been spent studying Chinook salmon, surprisingly few studies have used analytical methods to demonstrate differences in Chinook salmon migration pathways. The few studies that have shown differences in recovery patterns (e.g., Kondo et al., 1965; Hartt 1966; Aro, 1972a,b; Major et al. 1978; Hartt and Dell 1986; Healey, 1991; Waples et al., 2004) did not account for the geographical origins of the populations, had insufficient data for proper statistical analysis, or did not consider critical variables such as age at maturation and interannual variation. Weitkamp (2010) indicated that Chinook salmon have fairly stable distributions over time, but the issue of life history variation from the same stream of origin, and their consequential effect on migration patterns were not addressed in her analysis. Trudel et al. (2009) observed that immediately after ocean entry juvenile *O. tshawytscha* spend a large amount of time on the coastal shelf either migrating northwards towards Alaska or staying close to their natal rivers. This behavior was observed in both life history types, though later migration pathways were not presented in that study (Trudel et al., 2009), and are largely missing from the literature. Analysis of the migration patterns of stream- and ocean-type Chinook salmon is complicated by variations in behavior within each of the two types. Some ocean-type Chinook salmon from Oregon, for example, tend to be caught south of their river of origin whereas others are caught to the north (Nicholas and Hankin, 1989), and the spatial distribution of Chinook salmon from inland waters (e.g., Puget Sound and the Strait of Georgia) may differ from that of fish from the outer coast (Healey and Groot, 1987). Perhaps more fundamentally, the two life history types are segregated to some extent by latitude, which greatly affects the marine environments that they experience upon leaving freshwater. Southern populations (e.g., California, Oregon, and Washington) find suitable conditions for rearing in the coastal waters influenced by upwelling (Pearcy, 1992) but the waters directly offshore are too warm and unproductive for salmon. In contrast, more northerly populations can find cool, productive waters for rearing offshore and along the coast. Thus it is not clear whether the differences in catch distributions between the types reported by Healey (1983) and Waples et al. (2004) resulted from the juvenile life history or the region where the smolts entered the ocean.

The purpose of this study is to test the hypothesis that stream-type and ocean-type Chinook salmon differ in marine distributions. In this context, we use these terms in the original sense (i.e., Gilbert, 1913) to refer to juveniles migrating to sea in their second (stream-type) or first (ocean-type) year of life, without regard to the genetic lineages (e.g., Waples et al., 2004). We conducted a paired analysis of data from eight rivers where both types of Chinook salmon are produced and tagged. These included interior populations where genetic divergence between types is apparently ancestral, and coastal populations where divergence is apparently more recent (Waples et al., 2004). This study design allowed us to examine the factors influencing distribution (i.e., yearling vs sub-yearling, interior vs. coastal, and interactions between these factors). There is very limited fishing on the high seas but tagged salmon might be recovered in three general regions: 1) distant (primarily to the north) from the natal areas, along the coast, 2) coastal waters in the vicinity of the home river, and 3) the home river (either in the river or at the natal site itself). If the ocean distributions of salmon from a given river are similar for both types, no difference in relative

Table 1
Comparison of various life history and behavioral traits of characteristic North American ocean-type and stream-type Chinook salmon (based on Taylor, 1990a, 1990b; Healey, 1991).

Stream-type	Ocean-type
Dominant form in the northern part of the range in freshwater, and in the interior and at high elevation in the southern part of the ranges	Dominant form in the southern part of the range in freshwater, especially in lower elevation, coastal rivers
Juveniles reside in streams for about a year, including a winter, then migrate directly to the ocean in spring	Juveniles reside in streams for less than a year, migrating to sea shortly after emergence or after a few months in streams, with a broad distribution of migration timing from spring to early fall
Larger in size as smolts (70–140 mm long) but smaller for their age	Smaller in size as smolts (50–100 mm long) but larger for their age
Migrate rapidly through estuaries, primarily in the open water	Longer period of estuarine residence (as individuals and as populations), primarily in nearshore habitats
Spend 1–5 years at sea; typically 2–4, and fewer than ocean-type	Spend 1–5 years at sea; typically 2–4, and more than stream-type
Enter freshwater in spring and summer and spawn in the early fall (especially interior populations).	Enter freshwater in late summer or early fall and spawn in fall

recoveries among these three regions would be expected, although the distributions of salmon might vary among rivers. The predicted alternative, based on the hypothesis that stream-type Chinook salmon tend to migrate to offshore waters to rear (i.e., where they would not get caught), is that fewer stream-type fish would be recovered in distant coastal fisheries; instead they would be primarily recovered near the mouth of the natal river and in the river itself. We then used the data to determine whether there are differences in marine distribution among populations with the same life history type in different rivers. Based on our findings, we discuss the implications for the conservation of this species, and relate this to other conservation problems and solutions globally for similar depressed populations.

2. Materials and methods

Coded wire tags (CWT) have been used extensively to understand harvest and migration patterns of salmon since 1972 (Lapi et al., 1990; Johnson, 1990). We used CWT data from hatcheries that released both stream- and ocean-type Chinook salmon or paired nearby hatcheries in Washington, Oregon, Idaho, and British Columbia (Fig. 1; Table 2) for a total of eight pairs of populations. These populations represented eight geographical areas: i) the Siletz and Nestucca rivers on the Oregon coast, ii) Big Creek and the Willamette River on the Oregon side of the Columbia River, iii) the Cowlitz River on the Washington side of the Columbia River, iv) the Wenatchee River system (Methow and Okanagan rivers) and Icicle Creek for the Upper Columbia River, v) the Clearwater River and the lower Snake River for the Snake River, vi) the Chilliwack and Nicola rivers (for the lower Fraser River), vii) Dome Creek and the Shuswap River (upper Fraser river), and viii) the Kitsumkalum River and Kitimat Creek (Northern BC). We avoided hatcheries that modified the natural life history of the population (e.g., by extending the rearing of an ocean-type population to release them as yearlings). Recovery data were categorized by the different fishery areas along the west coast (Fig. 1).

The recovery rate (I) per fingerling or yearling released was estimated using equation (1).

$$I_{f,g,t} = \frac{\sum_{i=t+1}^{i=t+6} R_{i,f,g}}{S_{g,t}} \tag{1}$$

where R is the number of recoveries of the particular group (e.g., stream or ocean-type) at age (i), fishery (f), and at time (t), and S is the number of smolts released for that same group for a particular year (t). We first computed this rate for different ocean fishing areas (f) and then used these results in a Generalized Linear Model (GLM) form of Analysis of Variance to test for statistical significance of different design variables ("stock-type (G)", fishery (F) and year (T)) that might affect the recovery rate by area. We tested log-linear (eq. (2)) and log-transformed Poisson models (eq. (3)) as suggested by Green and McDonald (1987).

$$\ln\left(\frac{\sum_{i=t+1}^{i=t+6} R_{i,f,g}}{S_{g,t}}\right) = \ln\left(I_{f,g,t}\right) = \left(\alpha + \sum_{t=1}^{n} \beta_t T_t + \sum_{f=1}^{6} \gamma_f F_f\right) + \theta G_g + \sum_{f=1}^{6} \omega_{f,g}\left(G_g : F_f\right) + \varepsilon \tag{2}$$

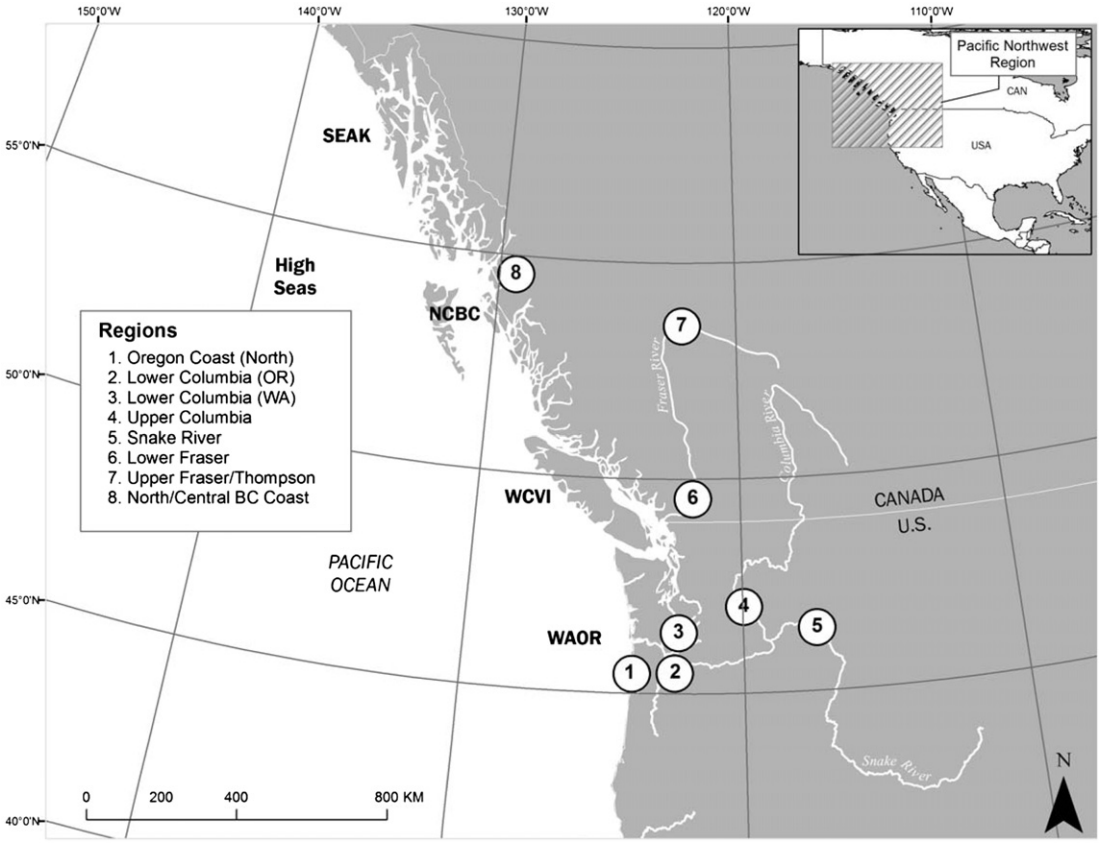


Fig. 1. Areas used in the analysis where stream-type and ocean-type Chinook salmon occur in close proximity.

Table 2
Locations where both stream-type and ocean-type stocks exist, and CWT data used to analyze the spatial distributions.

Region/location	Hatchery name	River	Brood years	Ocean or stream
Oregon Coast (North)	Cedar Creek	Nestucca River	77–81, 91–00	Stream
	Salmon River	Salmon (Siletz)	1977–2000	Ocean
Lower Columbia River,	Willamette River	Willamette River	1975–2000	Ocean
Oregon side	Big Creek/Stayton Ponds Hatchery	Lower Columbia River (OR side)	1976–2000	Stream
Lower Columbia River,	Cowlitz River	Cowlitz River	1979–2000	Ocean
Washington Side	Cowlitz River	Cowlitz River	1979–2000	Stream
Upper Columbia River	Wells	Upper Columbia River	1975–2000	Ocean
	Leavenworth National Fish Hatchery	Upper Columbia River/Icicle Creek/Yakama	1977–2000	Stream
Snake River (Idaho)	Lyons Ferry	Snake River	84–90, 92, 94, 98, 2000	Ocean
	Dworshak Hatchery	Clearwater River	1977–2000	Stream
Lower Fraser River	Spius Creek[a]	Nicola River/Bonaparte/Deadman/Coldwater	1984–2000	Stream
	Chilliwack Hatchery	Harrison River/Chilliwack	1981–2000	Ocean
Upper Fraser River	Shuswap River[a]	Shuswap River	1984–2000	Ocean
	Penny Creek[a]	Dome Creek/James Creek	1984–2000	Stream
Northern Coastal	Kitimat	Kitimat/Hersch Creek	1980–2000	Ocean
British Columbia	Terrace Hatchery	Kitsumakalum River	1979–2000	Stream

[a] Recovery data on escapement for brood years between 1997 and 2000 were applied to escapement data for all preceding years due to missing data prior to 1999 (Chuck Parken, Fisheries and Oceans Canada, Pacific Biological Station, 3190 Hammond Bay Road, Nanaimo, BC).

where $\sum R_{i,f,g}$ is the observed recovery of either stream-type or ocean-type Chinook (overall ages), T is the year effect design variable (between 10 and 25 years for any given stock), F is the fishery area design variable (six in our case: high seas, Alaska, northern British Columbia, the west coast of Vancouver Island, the coast of Washington and Oregon (combined), and terminal areas (i.e., river mouth and in-river locations)). All recoveries were summed separately by terminal areas, thus Columbia River stocks only sum up terminal Columbia River recoveries, and northern B.C. stocks only sum up northern B.C. terminal recoveries. S is the number of smolts released from the hatchery in the corresponding brood year, G is the group (stream- or ocean-type) effect variable, and $\omega_{f,g}$ is the parameter which describes the interaction between fishery, F and stock type, g. The first parameter (α) is the intercept, the second set of parameters, β_t, are year effects, which change the overall intercept by year (if significant), and the third set of parameters (γ_f) are fishery effects by area. To test if recoveries were different by stock type (G), a design variable is included for the stock type (stream-type = 1 and ocean-type = 0), and a parameter θ. This term can also change the intercept term in the model. Finally, the model included a stock-fishery interaction design variable which changed the intercept by fishery and stock type if significant. This type of analysis is very similar to a contingency table analysis which is a function of stock type, year and fishery, and possible interactions (Zar, 1995). Under the null hypothesis $\beta_t = \gamma_f = \theta = \omega_{f,g} = 0$. Under the alternative hypothesis β_t and/or γ_f and/or θ and/or $\omega_{f,g} \neq 0$. The most important terms to test for are θ and ω_f, to test whether there is any group effect (i.e., difference in recoveries between ocean- and stream-type groups) by area.

We also tested a Poisson Response log-transformed Model (Cormack and Skalski, 1992) to explain the large number of zeros seen in some strata (i.e., fishery recovery strata). Ocean recovery data are very sparse for some regions and locations. Consequently, the model to test these low numbers of recoveries used a Poisson error structure.

$$\mu_t = \exp\left(\left(\alpha + \sum_{t=1}^{n}\beta_t T_t + \sum_{f=1}^{6}\gamma_f F_f\right) + \left(\phi S_t(G_g)\right) + \theta G_g + \sum_{f=1}^{6}\omega_f G_g : F_f + \varepsilon\right) \quad (3)$$

where μ_t is the exponent of the linear predictor of the sum of recoveries in a strata (i.e.$\mu_t = \sum_{i=t+2}^{t+6} R_{i,g,f}$). Note taking logarithms on both sides of equation (3) makes it a linear model (eq. (4)).

$$LP = \left(\alpha + \sum_{t=1}^{n}\beta_t T_t + \sum_{f=1}^{6}\gamma_f F_f\right) + \kappa S_t + \left(\phi S_t(G_g)\right) + \theta G_g + \sum_{f=1}^{6}\omega_f G_g : F_f + \varepsilon \quad (4)$$

where LP is the Linear Predictor, S is the releases, T is the year effect, G is a stock effect, and F is the fishery (area) effect. This model was slightly different from the first model as it took into account a continuous variable (releases, S which is a covariate) to determine if there were any interactions by stock type (G). Such interactions would change the slope of the model based on "stock type". Other terms affect the outcome of the model as described in equation (3). That is, the model included a term for the stock type (stream or ocean-type, set as 0 for ocean-type and 1 for stream-type fish) which would change the intercept term in the model (the effect of a design/categorical variable changes the overall intercept of the model), and a stock-fishery interaction term which would change the intercept by fishery if significant. Thus, we now have 3 design (categorical) variables namely, stock type (G), fishery (F), year (T), and a continuous variable (covariate) releases (S).

This type of analysis is very similar to an analysis of covariance (ANCOVA) which is a function of stock type, year and fishery, and possible interactions (Zar, 1995). Under the null hypothesis: H_0: $\beta_t = \gamma_f = \theta = \kappa = \varphi = \omega_f = 0$. Under H_A: $\beta_t \neq 0$ and/or $\gamma_f \neq 0$ and/or $\theta \neq 0$, and/or $\varphi \neq 0$, and/or $\omega_f \neq 0$. In order to analyze an under-dispersed dataset, the F-test was used (Dr. Loveday Conquest, School of Fisheries and Aquatic Sciences, University of Washington, Seattle 98195, USA).

3. Results

3.1. Oregon coastal populations

Ocean- and stream-type fish from Oregon (Nestucca and Siletz rivers, represented by Cedar Creek and Salmon River hatchery programs) followed similar migration pathways (Fig. 2a) and both groups were recovered in all areas sampled. However, ocean-type

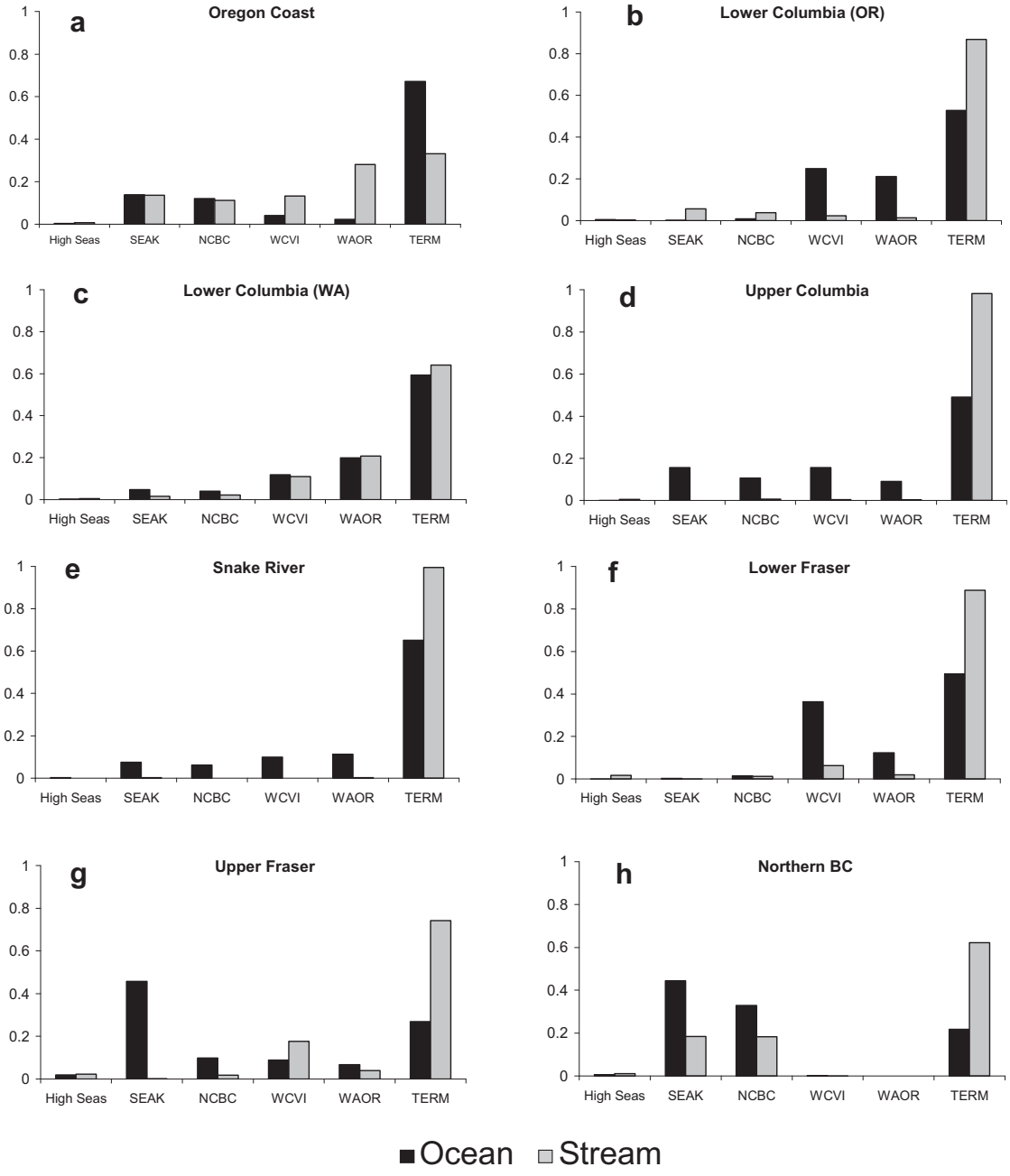


Fig. 2. Bar-plots indicating proportion of recoveries by ocean or stream-type Chinook salmon recoveries in the high seas, southeast Alaska (SEAK), northern coast of British Columbia (NCBC), west coast of Vancouver Island (WCVI), Washington and Oregon coasts (WAOR) and terminal areas aggregated by region.

fish were recovered at lower rates than stream-type fish in areas close to the natal region, in contrast to the prediction. This pattern indicates that these stream-type fish were more prevalent on the coastal shelf than ocean-type fish. When testing for differences using the two models (Tables 3and 4) all design (categorical) variables (year, area, and stock type), were significant as were group-area interactions, implying that these fish had different recovery rates in these fisheries. Regardless, the two types of Chinook salmon were recovered in similar areas of the ocean, and exhibited generally similar migrations across the shelf (Figs. 2a and 3).

3.2. Lower Columbia River (Oregon side)

Both ocean- and stream-type fish from the Willamette (Santiam and Klackamas hatcheries) and Big Creek (Stayton Ponds and Big Creek hatcheries) were recovered in the lower Columbia River system and in most ocean areas. However, ocean-type fish did not migrate as far north as stream-type (Figs. 2b and 4) from this region. Stream-type fish were recovered in small numbers in all areas, but primarily at terminal areas. Table 3 and Table 4 indicate that all design (categorical) variables were significant, though stock type-area interactions were marginally significant in the log-linear model ($p = 0.06$). The obvious differences in distribution patterns (Fig. 2b) between types supported Healey's hypothesis (1983), although some stream-type fish were recovered from the coastal shelf (Fig. 2b).

3.3. Lower Columbia River (Washington side)

Both ocean- and stream-type fish from the Cowlitz River hatchery were recovered in the lower Columbia River system and in

Table 3
ANOVAs showing results of equation (2) (log-linear model) applied to each of the particular areas.

	Df	Deviance	Resid. *Df*	Resid. Dev *F*	Pr(>*F*)	
Oregon Coast						
Null			227	3238.4		
t (Year)	24	316.6	203	2921.8	2.1	0.004138
f (Fishery/Area)	5	1342.8	198	1579	41.7	<2.2e-16
G (Stock type)	1	210.9	197	1368.1	32.8	3.94E-08
G:*f* (Stock:Area)	5	132.7	192	1235.4	4.1	0.001395
Lower Columbia (WA Side)						
Null			299	5322.8		
t (Year)	28	776.2	271	4546.6	4.4	<0.0001
f (Fishery/Area)	5	2715	266	1831.6	87.0	<0.0002
G (Stock type)	1	140.9	265	1690.7	22.6	<0.0003
G:*f* (Stock:Area)	5	67	260	1623.7	2.1	0.06
Snake						
Null			179	5509.6		
t (Year)	18	587.9	161	4921.7	6.8	2.71E-16
f (Fishery/Area)	5	1928.5	156	2993.3	80.4	<2.2e-16
G (Stock type)	1	1713.3	155	1280	356.9	<2.2e-16
G:*f* (Stock:Area)	5	560	150	720	23.3	<2.2e-16
Upper Fraser						
Null			191	3389.4		
t (Year)	16	151	175	3238.4	1.2	0.3092
f (Fishery/Area)	5	910.9	170	2327.5	22.3	<2.2e-16
G (Stock type)	1	312.2	169	2015.3	38.2	4.84E-09
G:*f* (Stock:Area)	5	675.9	164	1339.5	16.6	3.26E-13
Lower Columbia (OR Side)						
Null			311	6075.2		
t (Year)	26	760.6	285	5314.6	6.0	1.31E-15
f (Fishery/Area)	5	2134.3	280	3180.3	87.8	<2.2e-16
G (Stock type)	1	726.6	279	2453.6	149.5	<2.2e-16
G:*f* (Stock:Area)	5	1121.9	274	1331.8	46.2	<2.2e-16
Upper Columbia						
Null			269	8389.8		
t (Year)	25	1054.6	244	7335.2	8.4	<2.2e-16
f (Fishery/Area)	5	2548.6	239	4786.7	101.3	<2.2e-16
G (Stock type)	1	2297	238	2489.7	456.3	<2.2e-16
G:*f* (Stock:Area)	5	1316.8	233	1172.9	52.3	<2.2e-16
Lower Fraser						
Null			227	4566.9		
t (Year)	20	245.4	207	4321.5	1.6	0.04829
f (Fishery/Area)	5	2274.6	202	2046.9	60.5	<2.2e-16
G (Stock type)	1	358.5	201	1688.4	47.7	6.81E-11
G:*f* (Stock:Area)	5	215	196	1473.5	5.7	5.98E-05
North Central BC						
Null			251	7531.5		
t (Year)	21	140.9	230	7390.6	1.2	0.26559
f (Fishery/Area)	5	6084.5	225	1306.1	215.0	<2e-16
G (Stock type)	1	4.4	224	1301.7	0.8	0.37726
G:*f* (Stock:Area)	5	62.3	219	1239.4	2.2	0.05527

all ocean areas at similar rates except for the somewhat greater representation of ocean-type fish in northern areas (Fig. 2c). All design (categorical) variables were significant for this region (Tables 3and 4), marginally for the log-linear model ($p = 0.06$), and not for the Poisson model ($p = 0.31$). Migration pathways of the two types of Chinook salmon from this region overlapped (Fig. 4).

3.4. Upper Columbia River populations

Almost all recoveries for stream-type fish from the upper Columbia River (Wells hatchery representing the ocean-type fish in

Table 4
ANODEV showing results of equation (3) (Poisson response models) applied to each of the particular areas.

Variable	*Df*	Deviance	Resid *Df*	Resid Dev *F*	*p* > *F*	
Lower Columbia (WA Side)						
Null	299	729,919				
S (Releases)	1	73,328	298	656,591	33.3	<0.0001
t (Year)	28	339,081	270	317,511	10.3	<0.001
f (Fishery/Area)	5	254,047	265	63,463	212.2	<0.0001
G (Stock Type)	1	38,590	264	24,873	409.6	<0.0001
S:*G* (Releases:Stock)	1	1217	263	23,656	13.5	<0.001
f:*G* (Area:Stock)	5	723	258	22,933	1.6	0.31
Lower Columbia (OR Side)						
NULL	311	1,023,544				
S (Releases)	1	335,597	310	687,947	151.2	<0.0001
t (Year)	26	121,990	284	565,956	2.4	0.006
f (Fishery/area)	5	425,863	279	140,093	169.6	<0.0001
G (Stock type)	1	35,912	278	104,181	95.8	<0.0001
S:*G* (Releases:Stock)	1	2440	277	101,741	6.6	0.02
f:*G* (Area:Stock)	5	43,894	272	57,847	41.3	<0.0001
Upper Columbia						
Null	269	171,002				
S (Releases)	1	74,530	268	96,472	207.0	<0.0001
t (Year)	25	29,939	243	66,532	4.4	<0.0001
f (Fishery/area)	5	42,826	238	23,706	86.0	<0.0001
G (Stock type)	1	463	237	23,243	4.7	0.06
S:*G* (Releases:Stock)	1	2.70E-01	236	23,243	0.003	0.999
f:*G* (Area:Stock)	5	15,661	231	7581	95.4	<0.0001
Snake						
Null	179	48,028				
S (Releases)	1	20	178	48,008	0.1	0.999
t (Year)	18	10,826	160	37,182	2.6	0.002
f (Fishery/Area)	5	28,910	155	8272	108.3	<0.0001
G (Stock Type)	1	1808	154	6464	43.1	<0.0001
S:*G* (Releases:Stock)	1	3	153	6461	0.071	0.999
f:*G* (Area:Stock)	5	3464	148	2998	34.2	<0.0001
Lower Fraser						
Null	227	251,959				
S (Releases)	1	1481	226	250,478	1.3	0.51
t (Year)	20	23,708	206	226,770	1.1	0.7
f (Fishery/Area)	5	175,738	201	51,031	138.4	<0.0001
G (Stock Type)	1	10,755	200	40,276	53.4	<0.0001
S:*G* (Releases:Stock)	1	650	199	39,626	3.3	0.14
f:*G* (Area:Stock)	5	10,500	194	29,127	14.0	<0.0001
Upper Fraser						
Null	191	17,008.8				
S (Releases)	1	1512.7	190	15,496.1	18.5	<0.0001
t (Year)	16	1046	174	14,450.1	0.8	0.999
f (Fishery/Area)	5	7772.7	169	6677.4	39.3	<0.0001
G (Stock Type)	1	7.9	168	6669.5	0.2	0.999
S:*G* (Releases:Stock)	1	101.1	167	6568.4	2.6	0.21
f:*G* (Area:Stock)	5	3507.7	162	3060.8	37.1	<0.0001
Northern BC						
Null	251	45,415				
S (Releases)	1	760	250	44,655	4.3	0.07
t (Year)	21	5737	229	38,918	1.6	0.08
f (Fishery/area)	5	28,806	224	10,113	127.6	<0.0001
G (Stock type)	1	216	223	9897	4.9	0.06
S:*G* (Releases:Stock)	1	121	222	9776	2.7	0.2
f:*G* (Area:Stock)	5	4541	217	5235	37.6	<0.0001
Oregon Coast						
Null	227	182,548				
S (Releases)	1	58,132	226	124,416	105.6	<0.0001
t (Year)	24	21,053	202	103,363	1.7	0.05
f (Fishery/Area)	5	92,620	197	10,743	339.7	<0.0001
G (Stock Type)	1	1528	196	9215	32.5	<0.0001
S:*G* (Releases:Stock)	1	0.2226	195	9215	0.005	0.99
f:*G* (Area:Stock)	5	2250	190	6965	12.3	<0.0001

the Wenatchee, Okanagana and Methow systems, and Icicle Creek representing the Methow and Icicle creek systems) were in terminal areas, whereas ocean-type fish were recovered in all ocean areas from Alaska to the Columbia (Fig. 2d), indicating an off-shelf migration by stream-type and coastal shelf migrations by ocean-type Chinook salmon. All design (categorical) variables were significant although stock type was only marginally so ($p = 0.06$) in the Poisson model (Table 4). The stock-area interaction was highly significant (Table 3), implying differences in stock recovery patterns by area. Accordingly, stream-type migrations are shown in Fig. 5 as occurring off the shelf, and ocean-type fish are shown following the shelf.

3.5. Snake River populations

The Snake River fish (from Lyons Ferry hatchery representing the ocean-type fish and Dworshak Hatchery representing the stream-type fish) showed similar recovery and migration patterns as those from the upper Columbia River; few stream-type fish were recovered in ocean areas (Figs. 2e and 5). All design (categorical) variables were significant ($p < 0.05$) and stock type-area interactions were significant as well (Tables 3and 4).

3.6. Lower Fraser River populations

Lower Fraser River stream-type fish from the Nicola River (Fig. 2f) appeared to follow an open ocean migration pathway, appearing off the west coast of Vancouver Island and in the Strait of Juan de Fuca on their return migration. Ocean-type fish, in contrast from the Chilliwack River, followed a coastal shelf migration that split between the Strait of Georgia and the west coast of Vancouver Island on their return migration (Fig. 6). All design (categorical) variables were significant in the log-linear model (Table 3) but only area (categorical), stock type (categorical), and stock-area interaction were significant in the log-transformed Poisson model (Table 4).

3.7. Upper Fraser River populations

Stream-type fish from the Dome Creek hatchery showed an offshore migration pathway whereas ocean-type fish from the Shuswap River hatchery were closer to the coastal shelf (Fig. 2g). Most recoveries (75%) of stream-type fish were in terminal areas, with some ocean recoveries off Vancouver Island and Washington. All design (categorical) variables were significant in the log-linear model (Table 3) and stock type-areas interactions (Table 4) are significant in the Poisson model. Migration pathways (Fig. 6) indicate that stream-type fish migrate along the coastal shelf for some portion of their adult life but clearly differ from the coastal migration of the ocean-type fish.

3.8. Northern British Columbia populations

The migration pathways of northern British Columbia fish from the Kitimat and Kitsumkalum rivers (Terrace hatchery) overlapped significantly; both types were caught in southeast Alaska and northern B.C. (Figs. 2h and 3b). Stock-type was marginally significant in the log-linear model. Stock type-area interactions were not significant in the log-linear model (Table 3). The stock type was marginally significant ($p = 0.06$) and stock type-area interactions were significant in the Poisson response model (Table 4). The data suggest a migration along the shelf for ocean-type fish and a shelf, then open ocean then back to shelf pathway taken by the stream-type fish (Fig. 3b).

3.9. General patterns of ocean migration

When aggregated over space and time, distinct patterns in the data emerged (Fig. 2). There was a dramatic distinction between recovery patterns of stream- and ocean-type fish from the upper Columbia and Snake rivers (Fig. 2d and e), supporting the hypothesis that stream-type fish had an offshore distribution because higher proportions were recovered in terminal areas,

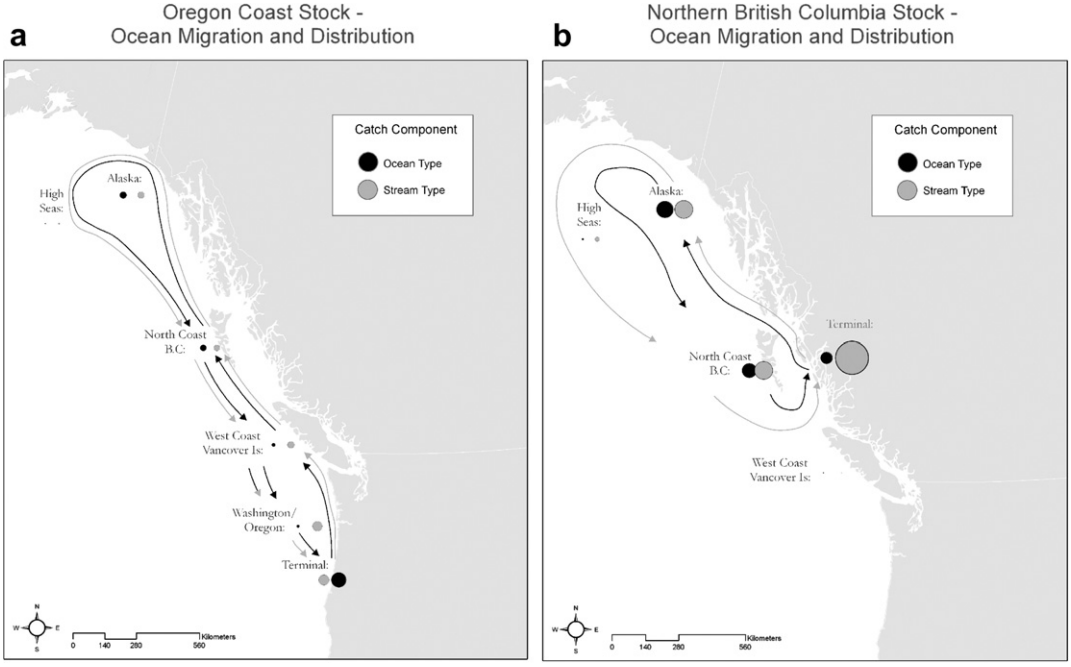


Fig. 3. Recovery patterns and migration pathways of Oregon coast (a) and Northern BC (b) Chinook salmon. Arrows show outward and return migrations. Circle sizes indicate proportion of overall recoveries in a particular area.

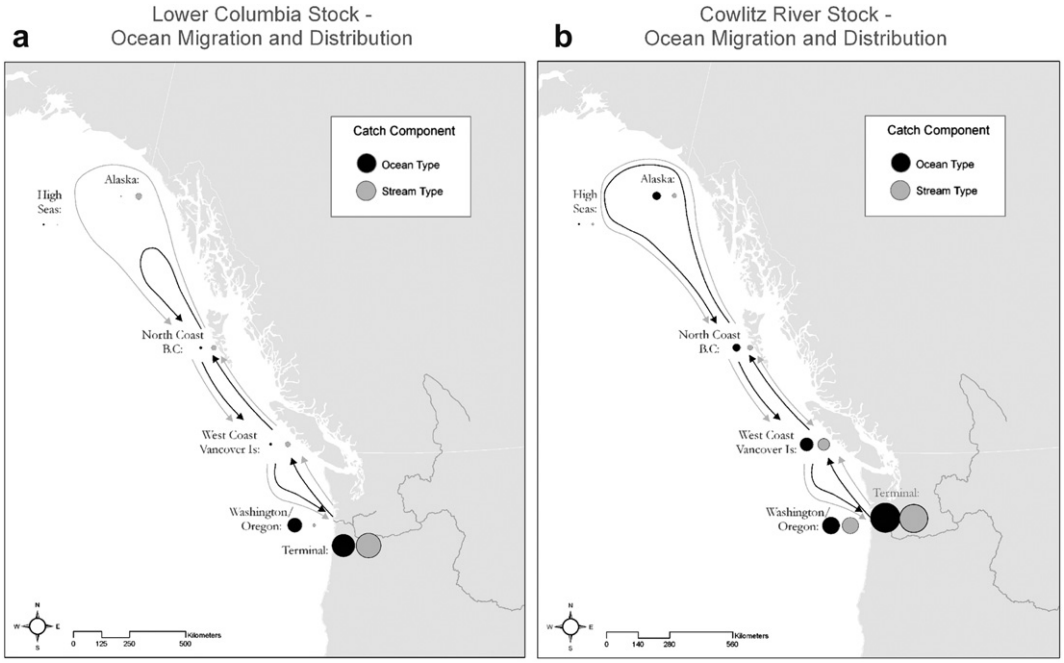


Fig. 4. Recovery patterns and migration pathways of lower Columbia River Chinook salmon (Oregon side, a, and Washington/Cowlitz side, b). Note arrows showing outward and return migrations. Circle sizes indicate proportion of overall recoveries in a particular area.

whereas ocean-type fish were more often caught on the continental shelf. However, the recovery distributions of Fraser River populations exhibited somewhat more overlap (Fig. 2f and g). Specifically, 25% of the recoveries from upper Fraser River stream-type fish were from the coast, as were 15% for lower Fraser River stream-type. In the lower Columbia River, both types were found in all recovery areas, indicating that the ocean distributions overlapped broadly (though this overlap was more evident in the Washington than Oregon populations). More overlap was also observed for the Oregon coast and northern B.C. populations than the upper Columbia River and Snake River populations.

In some cases (e.g., Oregon coast and Lower Columbia (Washington) River; Tables 3and 4) the area-stock interaction was not significant (Lower Columbia, Washington case), indicating that the main effect of the stock type (stream- or ocean-type) was significant with respect to the number of recoveries by area, but this effect was not different from the average effect by area (i.e. the ratio of the recovery patterns across areas by stream- or ocean-type remains the same across most regions). Thus, the insignificance of the stock-

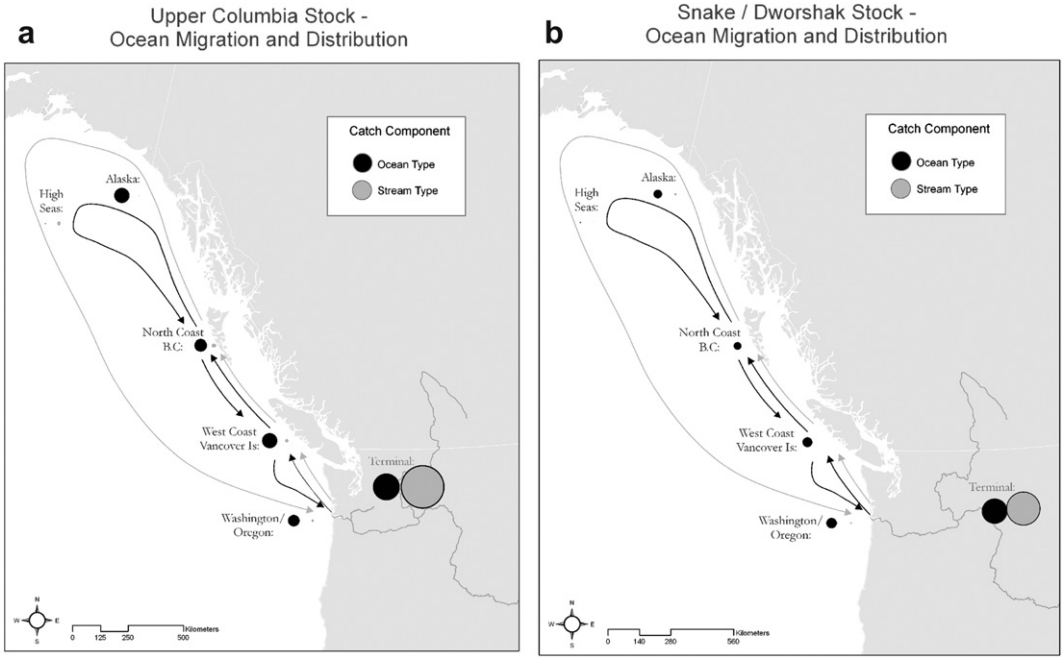


Fig. 5. Recovery patterns and migration pathways of upper Columbia River (a) and Snake River Chinook salmon (b). Arrows show outward and return migrations. Circle sizes indicate proportion of overall recoveries in a particular area.

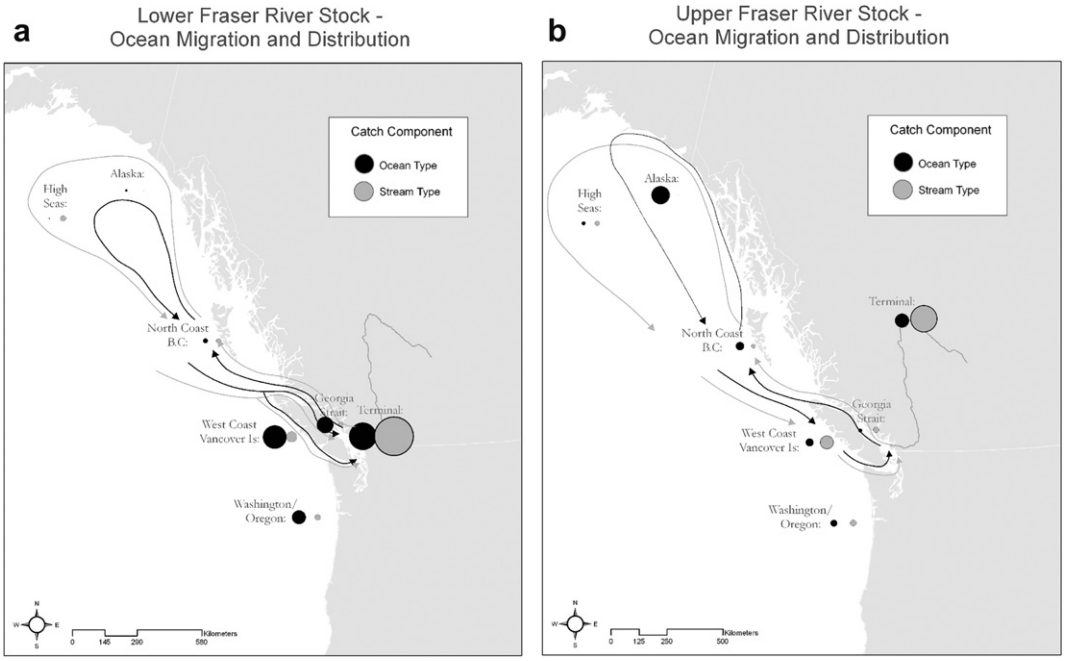


Fig. 6. Recovery patterns and migration pathways of lower (a) and upper (b) Fraser River Chinook salmon (arrows showing upstream and downstream migration). Circle sizes indicate proportion of overall recoveries in a particular area.

area interaction means that the ratio between the two stock types was constant over the regions where they were recovered. In the Oregon coast salmon, however, there was a difference though they were recovered in most areas, indicating a shelf migration for both ocean and stream-type Chinook salmon.

3.10. Model fit

Examination of the residual diagnostics of the fit for each of the regions and stock types revealed no apparent trend or bias for either the log-linear model or the log-transformed Poisson response model, although most of the diagnostics (quartile plots) suggested a concentration of data towards the center of the distribution and less at the tails. This result was expected, as the data were largely zeros with few recoveries in some years and fisheries, and numerous in other years and fisheries. In addition, the log-response model displayed some clear residual patterns in different clusters. These patterns did not appear when a Poisson response model was used. Regardless of the model chosen, in most cases the log-linear model indicated a significant stock and stock-area interaction effect, implying that there was a difference in recoveries by life history type that varied by region. In the case of the Poisson response model, the stock effect was significant in most cases, and the stock-area interaction effect was also significant in most cases ($p < 0.05$), implying that recovery patterns differed by life history type in all areas analyzed. The stock effects were not significant in the upper Fraser River, northern B.C., and the upper Columbia River in the Poisson response model. However, in all three cases the stock-area interaction was highly significant, indicating a difference in distribution patterns (Fig. 2). The only other case that showed no difference in area-stock recovery patterns once the main effect (i.e., life history type) was included in the model was the lower Columbia River (Cowlitz River) where the distributions of the two life history types overlapped considerably (Fig. 2).

When a simple test of proportions (Zar, 1995) was used on recoveries by year using release size as the denominator, proportions of recoveries by area were almost always significantly different by area across stock types. We do not to present these results as they are essentially the same results as those displayed in Tables 3and 4 using Generalized Linear Models (GLM).

3.11. Age and size differences in recovery patterns

Ocean-type fish enter the ocean at a smaller size and later date in the year than stream-type fish, and therefore might spend more years at sea prior to maturation. If so, they would have a higher chance of getting caught while feeding, decreasing the proportion of fish recovered in the terminal areas. To account for this possible bias, we analyzed the tag data in a different manner. We first assessed marine age of recoveries for the study populations and summarized them by age for the lower Columbia and Willamette rivers, the lower Fraser River, and Northern B.C. pairs of populations as examples (Table 5). Tables 3and 4 also indicated significant effects of the year the fish entered the ocean on distribution and maturation schedule. In general, the ocean-type fish spent more years at sea than the stream-type (see also Roni and Quinn, 1995). In some cases the age composition was sufficiently different as to affect catch opportunities. In the most extreme case, there was a very high proportion of ocean-age 0 fish in the Cowlitz River (lower Columbia River on the Washington side) hatchery's stream-type population and these fish (primarily males) would be essentially immune to ocean fishing. The other populations showed more moderate differences in age composition between types. This reduction in number of years spent at sea may reduce the vulnerability of stream-type fish to ocean fisheries. However, it does not seem sufficient to explain the extreme differences in catch patterns seen in some cases (e.g., upper Columbia River or Snake River), nor does it explain the differences in distribution between interior and coastal pairs of populations.

3.12. The effect of fishing on recovery patterns

To evaluate the possible effects of fishing effort in space and time, monthly catches were summarized for fisheries in southeast

Table 5
Recovery distributions by ocean-age (i.e., number of years at sea) and life history type for four pairs of populations.

Region	Life history type and fishery	Age 0	Age 1	Age 2	Age 3	Age 4	# of samples
Lower Columbia River, WA	Stream-type - ocean	0%	55%	42%	2%	0%	68,629
	Stream-type - terminal	37%	14%	34%	15%	0%	100,901
	Ocean-type - ocean	0%	8%	34%	49%	9%	7019
	Ocean-type - terminal	0%	2%	40%	52%	7%	5178
Lower Columbia River, OR	Stream-type - ocean	2%	21%	60%	17%	0%	10,915
	Stream-type - terminal	0%	2%	56%	41%	1%	120,300
	Ocean-type - ocean	0%	7%	75%	17%	1%	18,653
	Ocean-type - terminal	0%	8%	61%	30%	1%	15,313
Northern British Columbia	Stream-type - ocean	0%	4%	16%	60%	19%	7989
	Stream-type - terminal	0%	1%	9%	65%	25%	4252
	Ocean-type - ocean	0%	2%	25%	52%	20%	3500
	Ocean-type - terminal	0%	28%	16%	24%	32%	6692
Upper Fraser River	Stream-type - ocean	0%	5%	56%	37%	2%	517
	Stream-type - terminal	0%	2%	55%	42%	0%	1453
	Ocean-type - ocean	0%	1%	16%	76%	7%	9077
	Ocean-type - terminal	0%	8%	21%	67%	3%	2362

Alaska, Northern B.C., and the west coast of Vancouver Island (Fig. 7). The Alaskan fisheries operated year round and the Canadian fisheries operated all year except January and February but catches were low until summer. Thus the earlier migration from marine waters back to natal streams typical of the stream-type populations we examined would tend to reduce their vulnerability to coastal fisheries. However, as seen in Fig. 7, this is not the case. As was the case with age at maturity, the timing is not sufficient to explain the extreme differences in catch distribution between some pairs of populations, nor does it explain why some stream-type populations are much less vulnerable than others with similar timing.

4. Discussion

4.1. Distribution patterns according to juvenile life history type and origin

This study tested the hypotheses that i) juvenile life history type affects marine distribution (inferred from recovery data) regardless of where the fish entered the ocean and ii) there is a difference in marine distribution between life history types which varies among regions of origin. It is clear that the results presented here support both hypotheses, and indeed there was an interaction between the factors. In some cases (notably the Cowlitz River) ocean- and stream-type Chinook salmon were recovered at similar rates in all ocean areas, but in populations from the interior (the upper Columbia, Snake, and upper Fraser rivers) there was a clear difference in marine distributions between life history types. In some cases large differences in recovery patterns mirrored differences in life history, whereas in other cases the differences were small (though statistically significant); both types were recovered in all areas, but at different rates.

Given the large number of releases and replicates (about 200,000 tagged fish of each life history type on average over 10 years) and the large number of zeros observed, the Poisson model was probably the most appropriate way to analyze the data (Cormack and Skalski, 1992). It was also not surprising that differences were detected once the recoveries were aggregated by area. However, it is noteworthy that substantial numbers of stream-type fish were recovered in some of the same geographical areas where ocean-type fish were recovered. This result is not fully consistent with Healey's (1983) hypothesis that ocean-type fish have a more coastal distribution whereas stream-type fish are farther offshore, and that the two types have correspondingly different ocean migration pathways at maturation. This is apparent for coastal

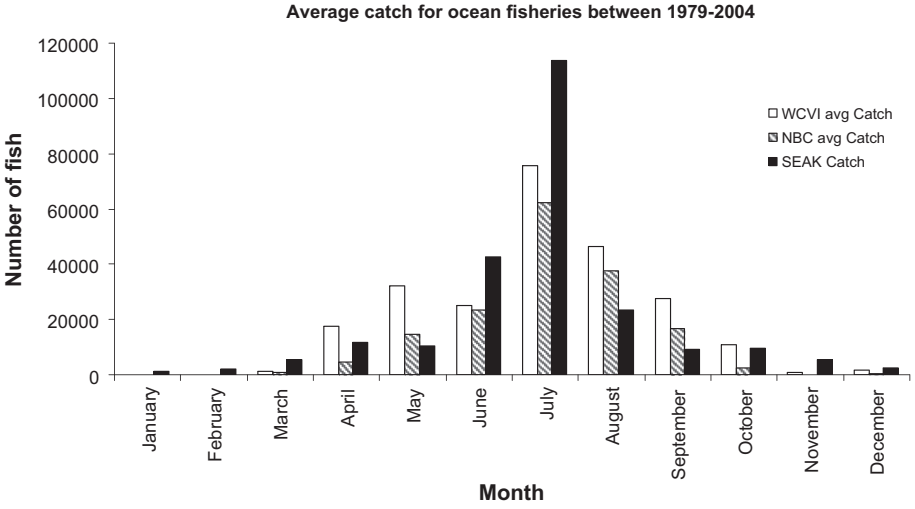


Fig. 7. Catch distributions by month for ocean troll fisheries in northern British Columbia, the west coast of Vancouver Island, and Southeast Alaska.

systems like the Oregon coast, and Northern BC (Fig. 3a and b) and the lower Columbia River systems (Fig. 4a and b). On the other hand, the data from populations in the continental interior (upper Fraser (Fig. 6d), upper Columbia (Fig. 5a), and Snake rivers Fig. 5b) (Fig. 5 and 6)) supported Healey's (1983) hypothesis). We had the benefit of vastly more data than Healey had access to because standardized surveys using coded wire tags were only implemented coast-wide in the late 1970s and early 1980s (Lapi et al., 1990; Johnson, 1990). Our extensive data and paired study design revealed a much more nuanced picture of the migratory behavior of these life history types than Healey (1983) or Waples et al. (2004) presented.

Both types of Chinook salmon from the lower Columbia River (Cowlitz River or Washington side), lower Fraser River, Oregon coast, and northern B.C. were recovered in broadly similar areas, though the patterns were not identical. Thus in these cases juvenile life history did not exert a fundamental control over distribution. Interestingly, ocean-type Chinook salmon from the lower Fraser River (Fig. 6) were evenly distributed between the Strait of Georgia and the west coast of Vancouver Island, but the ocean-type populations from the upper Fraser River, and stream-type Chinook salmon from both the lower and upper Fraser River, were rarely recovered from the Strait of Georgia. These latter populations appear to follow a shelf migration, followed by a wide turn out into the North Pacific Ocean, and back along the coastal shelf west of Vancouver Island when returning. In years of anomalous ocean conditions (e.g., El Niño in 1992) some of these fish were recovered off the northern coast of British Columbia, suggesting a shift in distribution in response to ocean conditions.

In contrast with Weitkamp's (2010) study, our results indicated that there is variation in stock recovery rates by year, and some inter-annual variation between stock recoveries in particular areas occurred over time. In addition, Weitkamp (2010) did not test for life history based differences in migration routes in the manner that we did. Our pair-wise treatment of similar life-histories by region and the consequential effect on region indicated some difference in recovery rates by region, though the data indicated that tags were recovered for the lower Columbia, lower Fraser, Oregon coast and Northern BC stocks in similar ocean areas for both stream and ocean-type fisheries. However, for upper Columbia and upper Fraser stocks there were clearer differences in missing ocean recoveries for stream-type stocks from most ocean areas.

4.2. Differences in migration patterns between upriver and downstream or coastal stocks

Why were the ocean migration patterns of the two juvenile life history types very different in pairs of populations from upriver regions but less distinct in the pairs from lower rivers or coastal areas? The degree of overlap in spawning may play a role by regulating interbreeding between forms. The lower river and coastal pairs of populations whose ocean distributions overlap also have overlapping spawning areas and periods (Narum et al., 2004; Waples et al., 2004). This overlap would allow some genetic exchange between the two life history types. In contrast, populations farther upstream seem to be more specialized in terms of their breeding location, timing, and life history. In the upper Columbia, Snake and upper Fraser rivers there is little or no overlap between the two types in their ocean recovery distributions (Figs. 5 and 6), and little overlap in spawning timing that would allow gene flow between the two types. This is consistent with the report by Waples et al. (2004) that these interior populations are genetically very distinct, and so in some sense may be the purest representatives of the "stream-type" if this term is taken to indicate lineage rather than merely the duration of juvenile residence in freshwater.

Brannon et al. (2004) hypothesized that the evolution of stream- and ocean-type Chinook salmon in the Columbia River system was mainly influenced by temperature. Stream-type juveniles emerge in cooler, more growth-limited environments in the upper parts of the river system and cannot grow large enough to overcome size-selective marine mortality in their first year of life, whereas ocean-type juveniles grow more quickly in the lower elevations. Ocean-type fish in the upper reaches of the river both spawn and rear in different areas than their stream-type counterparts, and move into larger expanses of water where they feed voraciously in May and June before reaching a threshold size for migrating to sea. Stream-type Chinook salmon migrate to sea as juveniles at a larger body size than their ocean-type conspecifics, they move more quickly through the estuary (Dawley et al., 1986; Ledgerwood et al., 1991), and seem to travel farther north initially. Some stream-type Chinook salmon from the Snake and upper Columbia rivers have been detected with acoustic tags near the Aleutian Islands, Alaska within a period of months after entering the ocean (Dr. David Welch, Kintama Research, Nanaimo, B.C., personal communication). These differences in marine migration and distribution patterns may be the key to understanding complex variation within and among years in survival of stream- and ocean-type populations, including some severely depleted ones.

4.3. The effect of age and size on recovery patterns

Despite our use of pair-wise comparisons between populations displaying the two life history types, interpretation of the underlying patterns and processes is nevertheless complicated by other factors. The ocean-type fish enter the ocean at a smaller size and later date in the year than stream-type fish, and if they compensate by spending more years at sea they will have more chances to get caught while feeding, decreasing the proportion of fish recovered in the terminal areas. The analysis to test for age effects however concluded that it does not seem sufficient to explain the extreme differences in catch patterns seen in some cases (e.g., upper Columbia River or Snake River), nor does it explain the differences in distribution between interior and coastal pairs of populations. From the conservation standpoint, it does not really matter why the stream-type are so much less vulnerable, though the process matters for understanding factors controlling migration.

4.4. The effect of fishing on recovery patterns

We examined the effects of fishing effort in space and time, monthly catches were summarized for fisheries in southeast Alaska, Northern B.C., and the west coast of Vancouver Island (Fig. 7). The Alaskan fisheries operated year round and the Canadian fisheries operated all year except January and February but catches were low until summer. Thus the earlier migration from marine waters to natal streams typical of the stream-type populations we examined would tend to reduce their vulnerability to coastal fisheries. Despite this effect, we concluded that timing is not sufficient to explain the extreme differences in catch distribution between some pairs of populations, nor does it explain why some stream-type populations are much less vulnerable than others with similar timing. The fact that stream-type fish were less frequently caught at sea remains, regardless of the reasons, though our efforts to understand their behavior are complicated by the interaction between duration of marine residence, homeward migration timing, and fishing patterns.

4.5. Is it valid to use hatchery fish to represent wild stocks and patterns?

Finally, the use of tagged hatchery fish to represent the behavior of wild populations is open to question. Both environmental and genetic factors affect behavior and life history, though Labelle et al. (1997) did not find consistent differences in distribution between wild and hatchery produced coho salmon. The populations that we examined did not have comparable wild-hatchery pairs but a comparison between wild and hatchery produced Upriver Bright Chinook salmon from the Columbia River indicated very similar distribution patterns (Appendix Table 1, TC-Chinook, 2011). This similarity is noteworthy because the wild fish from this population tended to be older than the hatchery population. Our analysis yielded average age composition in the catches from 1988 to 2001 in ocean troll fisheries from Oregon to Alaska as follows: wild: 5% age 3, 39% age 4 and 56% age 5; hatchery: 24% age 3, 55% age-4 and 21% age-5.

4.6. Management implications

These patterns of population-specific recoveries and, by implication, ocean distributions, are critical for management, as noted by Waples et al. (2004). For example, some stream-type Chinook salmon populations on the middle and upper Fraser River are of conservation concern because they are less productive than lower Fraser River stream-type and ocean-type populations (Parken et al., 2008). Based on Fig. 2g, 25% or more of the upper Fraser River population may be intercepted by ocean fisheries, especially those along the west coast of Vancouver Island and northern B.C. In the past it was assumed that the ocean catch of stream-type populations was small, so management directed its focus on in-river fisheries. However, the results demonstrated that ocean catch can be significant as well.

Conservation plans for other animal taxa with problematical spatial distributions have taken different forms. In cases such as dolphins, whose distributions overlap with tuna, area closures and mesh restrictions are common guidelines for preservation of these species in the oceanic environment (Hall, 1998). In other cases, like sea turtles, gear modifications (i.e., turtle exclusion devices) were designed to reduce by-catch without altering the distributions of shrimp and other trawl fisheries (Campbell et al., 2002; Shiode and Tokai, 2004). For some terrestrial species such as the snow leopard in the Himalayas (Mishra et al., 2003), and northern spotted owl (Murphy and Noon, 1992) in the Pacific Northwest, conservation corridors have been established for protection and rebuilding. Such approaches are lacking for endangered and depressed salmon, and although numerous approaches to protect the freshwater areas are being undertaken for these stocks, very limited actions in the oceanic environment occur due to the complications that closing fisheries would have for abundant and healthy populations mixed with the scarce or unproductive ones (Morishima and Henry, 2000). The key to recovery of some of these depressed stocks is looking at the entire life cycle and protecting all aspects of its life cycle that might cause the populations to decline. Thus, migration is a key attribute that needs to be addressed while developing management plans for threatened and endangered salmon, including marine distribution.

5. Conclusions

Chinook salmon have complex life history patterns and have evolved to colonize and persist in numerous areas throughout northern California, Oregon, Washington, British Columbia and Alaska, as well as Russia. Over the broad range of latitudes, elevations, and other features that characterize their habitats, growth rates of juveniles and the date and size of seaward migration differ greatly, with a mix of environmental and genetic regulation (Quinn, 2005). In some cases these differences in juvenile life history are linked to differences in marine distributions but in other cases they are not. Populations that are genetically more similar are also more similar in migration patterns. This hypothesis is consistent with studies (Narum et al., 2004; Waples et al., 2004) indicating considerable gene flow between the two life history types in the lower Columbia River but minimal gene flow between populations in the upper Columbia and upper Fraser rivers. Thus Healey's (1983) hypothesis is not uniformly supported in all areas, but it is partially true for the populations in the upper Columbia, Fraser, and Snake rivers. Consequently, creating migration corridors or time-area closures in fisheries that operate on both stock types will be important for conservation in the long run. Lastly, some of the stream-type salmon in the interior have experienced severe habitat modifications from human activities. Had they displayed similar migrations to ocean-type Chinook salmon from the same areas, the higher exploitation rates would have further increased their vulnerability and endangered status.

Acknowledgments

We thank David Graves (CRITFC) for his help in mapping some of the recovery data in Figures 3, 4, 5 and 6, and NOAA for funding this research. We thank CRITFC for supporting this research. Constructive comments were provided by Robert Francis, Ray Hilborn and Nate Mantua at the University of Washington in earlier versions of this manuscript.

Appendix

Table A1
Spatial distribution of wild and hatchery tagged Chinook salmon from the Upriver Bright population on the Columbia River, based on recoveries between 1988 and 2001.

Region and fishery	Hatchery	Wild
Alaska troll	11.7%	8.6%
Alaska net	0.5%	0.3%
Alaska sport	1.2%	1.3%
Northern B.C. troll	4.1%	3.8%
Central B.C. troll	0.1%	0.1%
Northern B.C. net	0.2%	0.4%
Northern B.C. sport	0.4%	0.5%
West coast Vancouver Island troll	3.4%	4.9%
Strait of Georgia troll and sport	0.1%	0.0%
Canadian net (area not specified)	0.2%	0.1%
Canadian sport (area not specified)	0.4%	0.4%
US troll	0.7%	0.7%
US sport	6.7%	6.3%
US net	18.2%	17.4%
Escapement	52.0%	55.3%

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